

Coding & Solution

Date	7 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask

Key Features Implemented	1. HDI Prediction using Linear Regression 2. User-friendly web interface 3. Predicts HDI from Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita 4. Displays HDI Score and Human Development Category 5. Input validation and real-time prediction
Pending / Incomplete Features	1. Cloud deployment 2. Interactive charts and visualizations 3. Live HDI data integration
Setup / Run Instructions	1. Install Python and required libraries. 2. Install dependencies using <code>pip install -r requirements.txt</code>. 3. Ensure HDI.pkl is in the project folder. 4. Run <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000</code> in a web browser. 6. Enter input values and click Predict.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	YES
2	Meaningful variable and function names are used	YES
3	Code includes comments / documentation where necessary	YES
4	Error handling is implemented for critical operations	YES
5	The application runs without critical errors	YES

6	Code is committed to a version control repository	YES
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Additional Notes / Comments:

The project successfully predicts the Human Development Index (HDI) using a Linear Regression machine learning model integrated with a Flask web application. Users can input Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita to receive an HDI score along with its corresponding development category. The application is user-friendly, responsive, and suitable for educational demonstrations.